

Project Planning Phase

Product Backlog · Sprint Planning · Stories · Story Points · Velocity · Burndown Chart

Date	15 February 2025
Team ID	HYD-METRO-2025
Project Name	Hyderabad Metro Rail – Digital Ticketing System (ServiceNow)
Maximum Marks	5 Marks

Key Agile Definitions

Term	Definition
Sprint	A fixed time-box (7 days) in which the team completes a defined set of tasks and delivers working increments.
Epic	A large feature too big for one sprint, broken into smaller Stories across multiple sprints.
Story (User Story)	Small deliverable task written as: 'As a [user], I can [action] so that [benefit].'
Story Point	Relative effort measure using Fibonacci scale: 1=Very Easy, 2=Easy, 3=Moderate, 5=Difficult, 8=Very Difficult.
Velocity	Average story points completed per sprint. Formula: Total Story Points ÷ Number of Sprints.
Burndown Chart	Graph of work remaining (story points) vs. time, showing whether the team is on track.

Story Points Scale (Fibonacci)

1 — Very Easy	2 — Easy	3 — Moderate	5 — Difficult	8 — Very Difficult
Update a label	Add a field	Write client script	Build a flow	Complex automation

Product Backlog & Sprint Schedule

The table below lists all 40 user stories across 6 sprints and 6 epics. SP = Story Points.

Sprint	Epic	USN	User Story / Task	SP	Priority	Member
Sprint-1	Table Creation & Data Setup	USN-001	Create Metro Station's Details table with all fields	3	High	Dev
		USN-002	Populate Metro Station Details with 10+ HYD stations	2	High	Dev
		USN-003	Create Metro Database custom table	3	High	Dev
		USN-004	Define field types, mandatory flags & field properties	2	High	Dev
		USN-005	Configure Reference fields linking Metro Station Details	3	High	Dev
		USN-006	Test table creation and field access in ServiceNow	2	High	Dev
		USN-007	Export update set for table configurations	3	Medium	Dev
			■ Sprint Total	18		
Sprint-2	Service Catalog Item & Variables	USN-008	Create 'Book A Metro Ticket' catalog item under Services	2	High	Dev
		USN-009	Configure 'Starting From' Reference variable	3	High	Dev
		USN-010	Configure 'Going To' Reference variable	3	High	Dev
		USN-011	Configure 'Type of Journey' Choice variable (Single/Return)	2	High	Dev
		USN-012	Configure 'No. of Passengers' String variable	2	High	Dev
		USN-013	Configure Mode of Payment Choice variable (5 options)	3	High	Dev
		USN-014	Configure 'Enter Payment Mode' conditional String variable	2	Medium	Dev
		USN-015	Add tooltips, example text, mandatory flags to all variables	3	Medium	Dev
			■ Sprint Total	20		
Sprint-3	Client Scripts & UI Policies	USN-016	Write Client Script: Fare Auto-Populate (OnChange)	5	High	Dev
		USN-017	Map all route-fare combinations (Ameerpet hub routes)	3	High	Dev
		USN-018	Write Client Script: Field Validator for No. of Passengers	2	High	Dev
		USN-019	Configure UI Policy: Mode of Payment = Others condition	3	High	Dev
		USN-020	Configure UI Policy Action: Show Enter Payment Mode field	2	High	Dev
		USN-021	Test all client scripts and UI policy in Service Portal	3	High	Dev
			■ Sprint Total	18		
Sprint-4	QR Code, Flow Designer & Automation	USN-022	Write OnSubmit Script: QR code generation via GlideModal	5	High	Dev

		USN-023	Configure GlideModal with external QR API page	3	High	Dev
		USN-024	Create Flow Designer flow: Metro Project	2	High	Dev
		USN-025	Configure Trigger: Service Catalog – Requested Item	2	High	Dev
		USN-026	Configure Flow Action 1: Get Catalog Variables	3	High	Dev
		USN-027	Configure Flow Action 2: Create Record in Metro Database	3	High	Dev
		USN-028	Test full flow end-to-end: catalog → DB record	2	High	Dev
			■ Sprint Total	20		
Sprint-5	ACLs, Business Rules & Portal	USN-029	Configure ACL rules on Metro Database table	3	High	Dev
		USN-030	Test ACL enforcement with non-admin user role	2	High	Dev
		USN-031	Write Business Rule: Generate Metro Ticket (on DB insert)	5	High	Dev
		USN-032	Create Service Portal page: Metro Ticket Page	3	Medium	Dev
		USN-033	Develop Metro QR Ticket widget on portal page	3	Medium	Dev
			■ Sprint Total	16		
Sprint-6	UAT, Documentation & Deployment	USN-034	Execute 10 UAT test cases (booking, fare, QR, ACL, flow)	3	High	Dev
		USN-035	Fix bugs identified during UAT	3	High	Dev
		USN-036	Export all update sets (tables, scripts, flow, ACL, portal)	2	High	Dev
		USN-037	Write Functional Specification Document (FSD)	3	Medium	Dev
		USN-038	Create project documentation for all phases	3	Medium	Dev
		USN-039	Final demonstration and stakeholder sign-off	2	High	Dev
		USN-040	Archive project and submit deliverables	2	Low	Dev
			■ Sprint Total	18		

Project Tracker, Velocity & Burndown Chart

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End (Planned)	SP Completed	Release Date (Actual)
Sprint-1	18	7 Days	10 Feb 2025	17 Feb 2025	18	17 Feb 2025
Sprint-2	20	7 Days	18 Feb 2025	25 Feb 2025	20	25 Feb 2025
Sprint-3	18	7 Days	26 Feb 2025	04 Mar 2025	18	04 Mar 2025
Sprint-4	20	7 Days	05 Mar 2025	12 Mar 2025	20	12 Mar 2025
Sprint-5	16	7 Days	13 Mar 2025	20 Mar 2025	16	20 Mar 2025
Sprint-6	18	7 Days	21 Mar 2025	28 Mar 2025	18	28 Mar 2025

Velocity Calculation

Sprint	Story Points	Cumulative Total
Sprint-1	18	18
Sprint-2	20	38
Sprint-3	18	56
Sprint-4	20	76
Sprint-5	16	92
Sprint-6	18	110
TOTAL	110	110
No. of Sprints	6	
VELOCITY = 110 ÷ 6	18 Story Points / Sprint	Team's Sprint Velocity

Velocity Formula

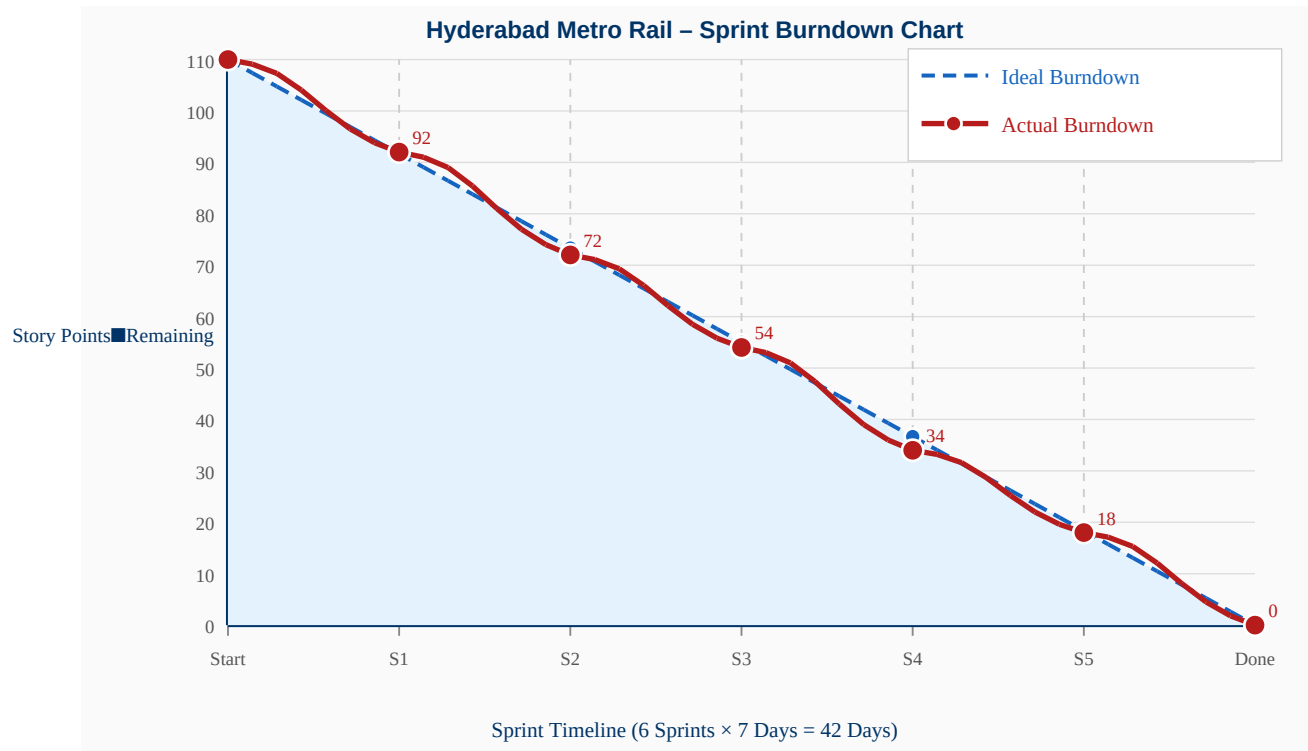
Velocity = Total Story Points ÷ Number of Sprints

Velocity = 110 ÷ 6 = 18 Story Points per Sprint

Average Velocity per day = Sprint Duration ÷ Velocity = 7 ÷ 18 ≈ 0.38 days per story point

Burndown Chart

The burndown chart shows story points remaining vs. sprint days. The ideal line represents perfect linear progress from start to finish. The actual line shows real sprint-by-sprint delivery across all 6 sprints of the Metro Rail Digital Ticketing project.



Burndown Data Table

Sprint	Days	Ideal SP Start	Ideal SP End	Actual SP Start	Actual SP End	On Track?
Sprint-1	0→7	110	92	110	92	✓ On Track
Sprint-2	7→14	92	73	92	72	✓ On Track
Sprint-3	14→21	73	55	72	54	✓ On Track
Sprint-4	21→28	55	37	54	34	✓ On Track
Sprint-5	28→35	37	18	34	18	✓ On Track
Sprint-6	35→42	18	0	18	0	✓ On Track
ALL	0→42	110	0	110	0	✓ Complete

Project Planning Summary

Metric	Value	Notes
Total Sprints	6	Each sprint = 7 days
Total Story Points	110	Across 40 user stories
Total User Stories	40	USN-001 to USN-040
Total Epics	6	One per sprint

Project Duration	42 Days	6 sprints × 7 days
Team Velocity	18 SP / Sprint	Total SP ÷ No. of Sprints
Avg Velocity/Day	2.6 SP / Day	Velocity ÷ Sprint Duration
All Sprints	100% Complete	All story points delivered on time